

Abdullah Sameh

DEVOPS ENGINEER

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SUMMARY

Data Science student with strong hands-on DevOps experience spanning Kubernetes orchestration, GitOps delivery, cloud infrastructure automation, and full-stack containerization. Built and deployed production-grade systems including a live federated social media platform on Kubernetes. Focused on cloud-agnostic, maintainable architectures and automation-first practices.

TECHNICAL SKILLS

Cloud	AWS (EC2, S3, IAM, VPC, EKS, CloudFront, CodeDeploy)
Observability	Prometheus, Grafana, Loki, Tempo, OpenTelemetry
Containers & Orchestration	Docker, Kubernetes, Helm, Kustomize
IaC	Terraform, Packer
CI/CD & GitOps	GitHub Actions, ArgoCD, CodeDeploy
Scripting	Bash, Python

PROJECTS

Mastodon — Production Kubernetes Deployment (Federated Social Media Platform) github.com/mastodon-SRE

- Building a Kubernetes deployment for a live federated social media platform (**Mastodon**), designed to run on **any online/cloud cluster** — AWS, Civo, or otherwise — with **autoscaling** across web, streaming, and Sidekiq, plus a companion **fediscoverer** service that scales the same way.
- Self-hosted, highly-available data layer: **PostgreSQL via CloudNativePG** (primary + replica), **Redis via DandyDeveloper's redis-ha chart** (Sentinel + HAProxy) for cache and Sidekiq queues, and **Elasticsearch via ECK** — no managed DB, cache, or search service dependency.
- Storage relies on whatever CSI the target cluster provides, and ingress is handled by **Traefik** — keeping the entire repo portable across cloud providers rather than tied to any one vendor's tooling.
- Cost-optimized node architecture: stateless app tier on **Spot instances**; stateful data services (PostgreSQL, Redis, Elasticsearch, MinIO) on **On-Demand nodes**.
- Full observability via a self-hosted **LGTM stack** (Grafana, Loki, Tempo, Mimir/Prometheus) running on the cluster itself, plus a basic **ChatOps** setup for day-to-day operational tasks and alerts.
- Two independent CI/CD pipelines via **GitHub Actions** — one for the app, one for infrastructure — both synced through **ArgoCD** GitOps, with TLS via cert-manager and secrets via Sealed Secrets.

MERN E-Commerce — AWS 3-Tier Production Architecture github.com/Passw0rd404/mern-ecommerce-DevOps-version

- Deployed a full-stack MERN app on AWS using a 3-tier architecture: **CloudFront** → **NLB** → **EC2 Auto Scaling Group** in a private subnet, with **ElastiCache** and **MongoDB Atlas** via PrivateLink.
- Two separate pipelines: frontend via S3 + CloudFront invalidation; backend via **CodeDeploy** blue/green. Infrastructure provisioned with **Terraform**, AMIs baked with **Packer**.
- Secured with OIDC keyless AWS auth, IaC scanning (TFLint, Terrascan, Checkov), and code quality gates (ESLint, Semgrep, OpenSSF Scorecard).

HiveBox — IoT Temperature API (Built from Scratch) github.com/Passw0rd404/HiveBox

- Wrote a **FastAPI** application from scratch aggregating real-time temperature data from global IoT sensors (OpenSenseMap API); containerized and deployed locally on **kind**.
- Integrated **Valkey** caching, Prometheus metrics, and a Grafana dashboard — full local observability stack without any cloud dependency.

HealthAI Labs — AI-Powered Health Platform (Built from Scratch) github.com/co-op-projects/HealthAI-Labs

- Built an AI health application from scratch integrating LLM APIs; containerized with **Docker Compose** — first experience shipping an AI-integrated service end to end.
- Practiced multi-service orchestration, environment variable management, and team collaboration with Git branching and pull request workflows.

EDUCATION

Bachelor's in Data Science, Faculty of Artificial Intelligence Expected 2028
Minoufia University — Egypt · GPA: 3.01

CERTIFICATIONS

AWS Solutions Architect Associate & Cloud Practitioner	Amazon Web Services View credential ↗
AWS DevOps — 90%	Cloud Native Base Camp View credential ↗
McKinsey Forward Program	McKinsey & Company View credential ↗